# Grade 12-Maths

## Unit 2

## Introduction to Calculus

### *Competency*

# *2.1 Introduction to derivatives*

## *Learning Outcomes*

**By the end of this section, you will be able to:**

- Understand rate of change of different quantities.
- Calculate rate of change of different quantities.
- Analyze the geometrical and mechanical meaning of derivative.
- Find average rates of change of functions between different points.
- Find the slope of a secant line and a tangent line to the graph of a function.
- Develop the concept of the derivative of a function at a point.
- Understand gradient of functions at a point.
- Find the derivative of simple functions using gradient method and at point

# *Definition to derivatives*

## *Learning Outcomes*

**By the end of this section, you will be able to:**

- State the definition of the derivative the gradient function.
- Explain what will happen when the gradient is zero.
- Find the derivative of some functions at a point.
- Find derivatives of power functions

# *Derivatives of combinations of functions*

## *Learning Outcomes*

**By the end of this section, you will be able to:**

- apply the sum, difference, product, and quotient formulae of differentiation functions.

# Maximum and minimum points

## *Competences*

**By the end of this sub-unit, you will be able to:**

- Find local maximum point and local minimum point of a given function on an interval
- Find Absolute maximum point and Absolute minimum point of a given function on an interval
- Write the equation of a tangent line and a normal line to a curve

# APPLICATIONS OF DERIVATIVE

## *Competences*

**By the end of this section, you will be able to:**

- Apply the concept of derivatives to solve problems on applications of derivative in different fields of study.

## Introduction to Integration

### *Competences*

**By the end of this section, you will be able to:**

- Define the *concept of integration* as the *reverse process of differentiation.*
- Understand the *geometric interpretation* of a *definite integral.*
- Find *area of a region bounded by a function and x-axis on a given interval*.
- Understand *definite integral*.
- Value the *real world contributions of having skills of derivatives and integrations*.
- Apply the *knowledge of integral calculus to solve real life mathematical problems*.

## Integration More on Areas and Application

***Competences***

**By the end of this section, you will be able to:**

- Find *area of a region bounded by a function and x-axis on a given interval*.
- Apply the *knowledge of integral calculus to solve real life mathematical problems*.
- Value the *real world contributions of having skills of derivatives and integrations*.

# THANK YOU!!!